BOUNDED GENERATION OF THE COHOMOLOGY ALONG POWERS OF THE IDEAL, WITH A CONTAINMENT FOR THE TRANSITION MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Finite generation of the graded module of cohomology along powers of the ideal. Same chapter spaces-cohomology.tex, label lemma-cohomology-powers-ideal-times-F (invoked at proof line 4287). It supplies finitely many elements $x_i \in H^p(X, I^{d_i}\mathcal{F})$ generating $\bigoplus_{n \geq 0} H^p(X, I^n\mathcal{F})$ over $B = \bigoplus_{n \geq 0} I^n$, which is what produces the constant $c = \max\{d_i\}$.
- The situation of formal functions: the algebraic space, the ideal sheaf and the module. Same chapter spaces-cohomology.tex, Situation \ref{situation-formal-functions} (statement line 4274). It fixes the ambient data X, I and $\mathcal{F}$ in which both conclusions are formulated, including the graded ring B acting on the graded module.

Source reference. Original lemma 08AW: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 4257–4257. Navigation pointer for \ref{lemma-cohomology-powers-ideal-times-F} used inside the counted span (line 4287); the argument that consumes it is retained in the span.

Source reference. Original reference 08AV: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 4243–4243. Navigation pointer for \ref{situation-formal-functions} used inside the counted span (line 4274); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 08AV. For every $p \geq 0$ there exists an integer $c \geq 0$ such that*

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *the multiplication map* $I^{n-c} \otimes H^p(X, I^c\mathcal{F}) \to H^p(X, I^n\mathcal{F})$ *is surjective for all* $n \geq c$, *and*
(2) *the image of* $H^p(X, I^{n+m}\mathcal{F}) \to H^p(X, I^n\mathcal{F})$ *is contained in the submodule* $I^{m-c}H^p(X, I^n\mathcal{F})$ *for all* $n \geq 0$, $m \geq c$.

Proof. By Lemma 08AW we can find $d_1, \ldots, d_t \geq 0$, and $x_i \in H^p(X, I^{d_i}\mathcal{F})$ such that $\bigoplus_{n\geq 0} H^p(X, I^n\mathcal{F})$ is generated by $x_1, \ldots, x_t$ over $B = \bigoplus_{n\geq 0} I^n$. Take $c = \max\{d_i\}$. It is clear that (1) holds. For (2) let $b = \max(0, n - c)$. Consider the commutative diagram of A-modules

$$\begin{array}{ccccc} I^{n+m-c-b} \otimes I^b \otimes H^p(X, I^c\mathcal{F}) & \longrightarrow & I^{n+m-c} \otimes H^p(X, I^c\mathcal{F}) & \longrightarrow & H^p(X, I^{n+m}\mathcal{F}) \\ \downarrow & & & & \downarrow \\ I^{n+m-c-b} \otimes H^p(X, I^n\mathcal{F}) & & \longrightarrow & & H^p(X, I^n\mathcal{F}) \end{array}$$

By part (1) of the lemma the composition of the horizontal arrows is surjective if $n + m \geq c$. On the other hand, it is clear that $n + m - c - b \geq m - c$. Hence part (2). □